

Loci of complex numbers

Key Points

The loci of

$$|z - w| = r$$

is the set of complex numbers such that the distance from w is r (i.e a circle of radius r centered at w)

The loci of

$$\text{Arg}(z - w) = \theta$$

Is the set of complex number centred at w which all have the same argument (i.e a half-line starting at w with an angle θ to the horizon)

Basic Skills

Q1. Work out the modulus and argument of

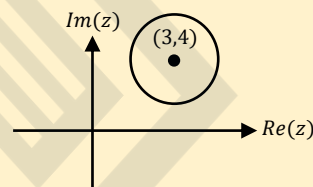
- a) $2 + 3i$
- b) $3i - 2$
- c) $-(1 + i)$

Q2. Draw the following loci in the complex plane

- a) $|z| = 4$
- b) $\text{Arg}(z) = \frac{\pi}{4}$
- c) $|z - (2 + 3i)| = 4$
- d) $\text{Arg}(z - (2 + 3i)) = \frac{\pi}{4}$

Competency Skills

Q3. Someone has drawn the loci of $|z - 3 + 4i| = 10$, point out two things wrong with their diagram



Q4. Draw the following loci

- a) $|z - 5i + 2| = 2$
- b) $\text{Arg}(z - 5) = \frac{2\pi}{3}$
- c) $|z + 2i + 1| = 1$
- d) $\text{Arg}(z + 3i - 5) = 1$

Q5. Draw on the complex plane the set of points such that $|z| > 1$

Q6. Draw on the complex plane the set of points such that $-\frac{\pi}{3} \leq \text{Arg}(z) \leq \frac{\pi}{5}$

Q7. Draw the following loci

- a) $1 < |z - 5i + 2| \leq 2$
- b) $\text{Arg}(z - 5) < \frac{2\pi}{3}$
- c) $|z^2| > 4$
- d) $\text{Arg}(z + 3i - 5) > 1$

Advanced Skills

Q8. a) Explain why $|z - w| = |w - z|$ using $z = x + yi, w = u + vi$

b) Explain why $|z - w| = |w - z|$ geometrically

c) Hence draw the loci for $|i - 6 - z| = 5$

Q9. By using $z = x + yi$ convert the loci of $|z - (a + bi)| = r$ into $(x - a)^2 + (y - b)^2 = r^2$

Q10. a) Given that $\text{Arg}\left(\frac{z}{w}\right) = \text{Arg}(z) - \text{Arg}(w)$ show that $\text{Arg}\left(\frac{z-3}{z+2}\right) = \frac{\pi}{4}$ is equivalent to $\theta - \phi = \frac{\pi}{4}$ where $\theta = \text{Arg}(z - 3), \phi = \text{Arg}(z + 2)$

b) By plotting $\theta = \text{Arg}(z - 3), \phi = \text{Arg}(z + 2)$ where $\theta - \phi = \frac{\pi}{4}$ find the loci for $\text{Arg}\left(\frac{z-3}{z+2}\right) = \frac{\pi}{4}$

Extension

Q11. By drawing the set of loci for when z satisfies $|2 + 3i - z|$ and $\text{Arg}(z - 2i) = \frac{\pi}{3}$ find the values of z

Hint: Convert the problem to cartesian coordinates after you've drawn it